WHERE Filter in MySQL

WHERE is one of the most important parts of SQL.
It is used to **filter rows** from a table based on conditions.

Think of it like:

"Give me only rows that match my rules."

Basic Syntax

SELECT column_names

FROM table_name

WHERE condition;

Example:

SELECT *

FROM employees

WHERE department_id = 101;

👉 Returns only employees from department 101.

Real-Life Analogy

Imagine a company with 10,000 employees.

Without WHERE:

SELECT * FROM employees;

👉 Gives all employees.

With WHERE:

SELECT * FROM employees WHERE salary > 50000;

👉 Gives only employees earning above 50k.

So WHERE saves time, data, and effort.

Internal Execution Order

When MySQL runs:

SELECT *

FROM employees

WHERE salary > 50000;

Logical order:

1. FROM employees
2. Read rows
3. Apply WHERE filter
4. Return matching rows

So **WHERE happens before SELECT output is shown**.

All Major Cases of WHERE

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1. Equality Filter =

SELECT * FROM employees

WHERE department_id = 101;

Only rows where department_id is exactly 101.

---

2. Not Equal != or <>

SELECT * FROM employees

WHERE department_id <> 101;

All rows except department 101.

---

3. Greater / Less Than

WHERE salary > 50000

WHERE salary < 30000

WHERE salary >= 45000

WHERE salary <= 70000

Used for salary, age, price, marks etc.

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4. Multiple Conditions using AND

SELECT *

FROM employees

WHERE department_id = 101

AND salary > 40000;

Both conditions must be true.

👉 Dept 101 employees with salary > 40k

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5. Multiple Conditions using OR

```
SELECT *
FROM employees
WHERE department_id = 101
OR department_id = 102;
```

Any one condition true.

👉 Employees from 101 or 102

---

6. Use Parentheses (Very Important)

```
SELECT *
FROM employees
WHERE (department_id = 101 OR department_id = 102)
AND salary > 50000;
```

Without brackets results may differ.

---

7. BETWEEN Range Filter

```
SELECT *
FROM employees
WHERE salary BETWEEN 30000 AND 60000;
```

Same as:

```
salary >= 30000 AND salary <= 60000
```

---

8. IN List Filter

```
SELECT *
FROM employees
WHERE department_id IN (101,102,103);
```

Cleaner than multiple ORs.

Equivalent:

department_id=101 OR department_id=102 OR department_id=103

---

9. NOT IN

SELECT *

FROM employees

WHERE department_id NOT IN (101,102);

All except those departments.

---

10. LIKE Pattern Search

Used for text matching.

WHERE first_name LIKE 'J%'

Starts with J

WHERE first_name LIKE '%n'

Ends with n

WHERE first_name LIKE '%oh%'

Contains oh

WHERE phone_number LIKE '98%'

Starts with 98

Wildcards

Symbol	Meaning
%	Any number of characters
_	One single character

Example:

WHERE first_name LIKE '_a%'

2nd letter is a

---

11. NULL Checks

Wrong:

WHERE manager_id = NULL

Correct:

WHERE manager_id IS NULL

No manager assigned.

WHERE manager_id IS NOT NULL

Manager exists.

---

12. Date Filtering

WHERE hire_date = '2024-01-01'

WHERE hire_date > '2023-12-31'

WHERE hire_date BETWEEN '2024-01-01' AND '2024-12-31'

---

13. Arithmetic in WHERE

WHERE salary + 5000 > 60000

Works, but slower than:

WHERE salary > 55000

Because indexed column transformed.

---

14. Functions in WHERE

WHERE YEAR(hire_date)=2024

Better:

WHERE hire_date >= '2024-01-01'

AND hire_date < '2025-01-01'

(Faster with index)

---

15. Subquery in WHERE

```
SELECT *
FROM employees
WHERE department_id IN
(
  SELECT department_id
  FROM departments
```

```
  WHERE location='Bangalore'
);
```

Very common in real work.

---

16. EXISTS

```
SELECT *
FROM employees e
WHERE EXISTS
(
  SELECT 1
  FROM bonus b
  WHERE b.employee_id = e.employee_id
);
```

Employees having bonus records.

WHERE with UPDATE

```
UPDATE employees
SET salary = salary + 5000
WHERE department_id = 101;
```

Only dept 101 updated.

⚠️ Without WHERE all rows updated.

WHERE with DELETE

```
DELETE FROM employees
WHERE employee_id = 5;
```

Deletes one row.

⚠️ Without WHERE:

```
DELETE FROM employees;
```

Deletes all rows.

WHERE vs HAVING

WHERE	HAVING
Filters rows before grouping	Filters groups after GROUP BY
Uses raw rows	Uses aggregate results

Example:

```
SELECT department_id, COUNT(*)
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5;
```

Important Operator Priority

MySQL checks:

1. ()
2. NOT
3. AND
4. OR

Example:

```
WHERE dept=101 OR dept=102 AND job='Manager'
```

Means:

```
WHERE dept=101 OR (dept=102 AND job='Manager')
```

Performance Tips (Real Industry)

Good

```
WHERE department_id = 101
```

(Index friendly)

Bad

```
WHERE department_id + 0 = 101
```

Bad

```
WHERE YEAR(hire_date)=2024
```

Better

```
WHERE hire_date >= '2024-01-01'
```

AND hire_date < '2025-01-01'

Common Beginner Mistakes

1. Forget WHERE in DELETE

DELETE FROM employees;

Dangerous.

2. Using = NULL

Wrong.

3. AND / OR confusion

Use parentheses.

4. Case sensitivity confusion in strings depends on collation.

Real Company Use Cases

HR

WHERE salary > 100000

Ecommerce

WHERE price < 500

AND stock > 0

Banking

WHERE transaction_date = CURDATE()

Data Engineering

WHERE created_at >= NOW() - INTERVAL 1 DAY

Example with Your Table

SELECT *

FROM employees

WHERE department_id = 101

AND salary > 40000

AND hire_date >= '2024-01-01';

Meaning:

- Dept 101

- Salary above 40k
- Joined in 2024 or later

Internal MySQL Working

MySQL may:

1. Use index to find matching rows quickly
2. Scan table if no index
3. Evaluate condition row by row
4. Return matched rows

Golden Rule

Use WHERE to reduce rows as early as possible.

Less rows = faster query.

Interview Definition

WHERE clause is used to filter rows from a table based on specified conditions before grouping or returning results.

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10 Practice Queries

1. employees with salary > 50000

2. employees in dept 101

3. employees in 101 or 102

4. employees not in 103

5. joined after 2024

6. first_name starts with A

7. manager_id is null

8. salary between 30k and 60k

9. job_title = 'Manager'

10. dept 101 and salary > 40k

MySQL Conditions

These are used inside WHERE clause to filter rows.

SELECT * FROM employees WHERE condition;

---

1. AND

AND means **all conditions must be true**

Example

SELECT * FROM employees

WHERE department_id = 101

AND salary > 40000;

Meaning:

Give employees:

- in department 101
- and salary above 40000

Both must match.

Simple Logic

Condition 1	Condition 2	Result
True	True	✅
True	False	❌
False	True	❌
False	False	❌

---

2. OR

OR means **any one condition true is enough**

Example

SELECT * FROM employees

WHERE department_id = 101

OR department_id = 102;

Meaning:

Employees from:

- department 101
  or
- department 102

Logic

Condition 1	Condition 2	Result
True	True	✅
True	False	✅
False	True	✅
False	False	❌

---

3. NOT

NOT means reverse / opposite

Example

SELECT * FROM employees

WHERE NOT department_id = 101;

Meaning:

All employees **except** department 101

Same as:

WHERE department_id <> 101

---

4. BETWEEN

Used for ranges.

Example

SELECT * FROM employees

WHERE salary BETWEEN 30000 AND 50000;

Meaning:

Salary from 30000 to 50000 inclusive.

Same as:

WHERE salary >= 30000 AND salary <= 50000

Date Example

WHERE hire_date BETWEEN '2024-01-01' AND '2024-12-31'

---

5. IS NULL

Checks empty / no value.

Example

SELECT * FROM employees

WHERE manager_id IS NULL;

Meaning:

Employees who have no manager assigned.

---

6. IS NOT NULL

Checks value exists.

Example

SELECT * FROM employees

WHERE manager_id IS NOT NULL;

Meaning:

Employees who have manager assigned.

Why not use = NULL ?

Wrong:

WHERE manager_id = NULL

Correct:

WHERE manager_id IS NULL

Because NULL means unknown / no value.

Combined Example

SELECT * FROM employees

WHERE department_id = 101

AND salary BETWEEN 30000 AND 60000

AND manager_id IS NOT NULL;

Meaning:

Give employees:

- Dept 101
- Salary 30k to 60k
- Have manager

Easy Memory Trick

Keyword	Meaning
AND	All true
OR	Any true
NOT	Opposite
BETWEEN	Range
IS NULL	No value
IS NOT NULL	Has value

Real Example from Daily Life

Want pizza order:

Veg AND Large

Means both needed.

Veg OR Paneer

Any one okay.

NOT spicy

No spicy.

Price BETWEEN 200 AND 500

Range.

Practice Queries

1. salary > 40000 AND department_id=101

2. department_id=101 OR department_id=102

3. NOT department_id=103

4. salary BETWEEN 25000 AND 45000

5. manager_id IS NULL

6. email IS NOT NULL

Important Tip

When mixing AND + OR, use brackets.

WHERE (department_id=101 OR department_id=102)

AND salary > 40000